

Pull Request: Merge development -> main

Summary

This PR merges the completed Task 1 (EDA) and Task 2 (Quantitative Analysis) work from the development branch into main. This represents a major milestone of the Financial News Sentiment Analysis project, establishing comprehensive exploratory data analysis and quantitative analysis frameworks.

Type of Change

- Major feature release (Task 1 & Task 2 completion)

Overview

This merge brings the complete Task 1 EDA and Task 2 Quantitative Analysis implementations to the main branch, including:

- 7 comprehensive EDA notebooks covering all aspects of exploratory analysis

What's Included

Task 1: Exploratory Data Analysis (7 notebooks)

1. 01_Data>Loading_and_Setup.ipynb - Data loading, quality checks, cleaning, preprocessing

Task 2: Quantitative Analysis (1 notebook)

8. 07_Quantitative_Analysis.ipynb - Technical indicators, financial metrics, trading signals

Infrastructure

- notebooks/utlis.py: Reusable utility functions (398 lines)

Visualizations

- 27 high-resolution (300 DPI) PNG files organized in notebooks/figures/

Key Achievements

Data Analysis Coverage

- [DONE] Exploratory Data Analysis: Complete analysis of 1.4+ million financial news headlines
- [DONE] Text Analysis: NLP techniques including keyword extraction, event detection, and topic modeling
- [DONE] Time Series Analysis: Comprehensive temporal pattern analysis across multiple time scales
- [DONE] Publisher Analysis: Deep dive into publisher characteristics and specialization
- [DONE] Cross-Analysis: Multi-dimensional insights combining stocks, publishers, and events
- [DONE] Quantitative Analysis: Technical indicators and financial metrics for 6 stocks
- [DONE] Executive Summary: Consolidated findings and recommendations

Technical Accomplishments

- Established modular notebook structure for maintainability

Dataset Insights

- Analyzed 1.4+ million financial news headlines

Quality Assurance

- [DONE] All notebooks tested and executed successfully

Breaking Changes

None - This is the first major feature addition to main branch from development.

Migration Guide

N/A - No migration required. This is initial implementation.

Dependencies

All dependencies listed in requirements.txt. Key libraries:

- pandas, numpy

Testing

- All notebooks executed successfully

Documentation

- Comprehensive notebook structure documentation

Files Changed

- 8 new notebook files (7 EDA + 1 quantitative analysis)

Performance Considerations

- Processed data saved to pickle for efficient reuse

Security Considerations

- No sensitive data exposed

Next Steps After Merge

1. Begin Task 3: Correlation between News and Stock Movement

Related Work

- Task 1: Git and GitHub Setup & EDA (Completed)

Approval Checklist

- Code follows project structure

Status: [DONE] Ready for merge to main

Impact: High - Establishes complete foundation for sentiment correlation analysis

Risk: Low - Well-tested, documented, and structured implementation